

Poly(ionic liquid)s as efficient and recyclable catalysts for methanolysis of PET

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ABSTRACT

The methanolysis of polyethylene terephthalate (PET) was investigated using poly(ionic liquid)s (PILs) synthesized with 1-vinyl-3-ethylimidazole acetates ([VEIm]Ac) and acrylic acid metal salts, and the activities of the PILs were screened. The effects of reaction temperature, reaction time, methanol amount, catalyst amount and PET sources on the methanolysis of PET were investigated using the optimized catalyst PIL-Zn²⁺. Under the optimal conditions the conversion of PET was 100% and the yield of dimethyl terephthalate (DMT) was 89.1%. In addition, the catalyst has very good applicability for methanolysis of different PET raw materials. More importantly, the PIL-Zn²⁺ catalyst could be recycled by simple filtration after the reaction, and could be reused six times without significant decrease in PET conversion and DMT yield. The catalysts were analyzed by TGA, FT-IR, XRD and ICP before and after recycling, the results showed no change in the structure of the catalyst, reflecting the good stability of the catalyst. The reaction kinetics of PIL-Zn²⁺ was first-order with an activation energy of 107.6 kJ/mol. This work mainly provides a way to design efficient, stable and recyclable catalysts for PET methanolysis.

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1. Introduction

Polyethylene terephthalate (PET) is a semi-crystalline thermoplastic polyester with high strength, transparency and safety [1]. Because of its light weight, nontoxic, good stability and low price, it is widely used in textiles, electronic products, beverage containers and food packaging [2,3]. It is one of the most widely used plastics in the world. According to statistics, the global annual production of PET exceeds 70 million tons, of which 66% is used for fibers and 30% for packaging, but only 14% of the waste PET is recycled [4]. PET has a stable chemical structure and is difficult to degrade in the natural environment. The existence of large amounts

of waste plastics causes enormous pressure on the environment and economy, thus, the sustainable recycling of waste PET has become a global concern [5].

Currently, the main recycling methods for PET are physical strategies, chemical recycling and biodegradation. Among these methods, chemical recycling, which converts waste PET into valuable chemicals, has received a lot of attention from the "sustainable society" and "green chemistry" viewpoints [6]. The main routes for chemical recovery of PET are pyrolysis [7], hydrolysis [8,9], methanolysis [10,11], glycolysis [12,13], other alcoholysis [14–16] and aminolysis/ammonolysis [17–19]. Among them, glycolysis of PET has been widely studied, but it suffers from high reaction temperature, difficult in catalyst recovery and product purification. In contrast, catalytic methanolysis can effectively avoid the generation of colored by-products during the degradation process. Even if the raw material used is colored waste PET, it will not affect the methanolysis and subsequent decolorization and purification of DMT. The degraded product DMT can also be used to produce PET, PETG and other high value-added products, which has a good prospect. Low-temperature catalytic methanolysis of PET is one of

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the most promising strategies to solve the above problems. The catalysts used for PET methanolysis can interact with the oxygen of PET carbonyl group to promote the penetration of methanol, which helps to reduce the reaction temperature and save energy [10].

Presently, the commonly used catalysts for PET degradation mainly include multiphase catalysts of metal oxides (ZnO, Co₃O₄, Mn₃O₄, ZnMn₂O₄, etc.) [20], homogeneous catalysts of metal acetates (Zn, Mn, Co, etc.) [21,22] and ionic liquids ([Bmim][Cl], [Bmim][AlCl₄], [Deim][Zn(OAc)₃], etc.) [23,24]. Among them, zinc-based catalysts show higher activity for PET depolymerization [25]. However, heterogeneous catalysts suffer from high reaction temperature and low efficiency, while homogeneous catalysts suffer from difficulty in recovery and residue in the product. Therefore, it is necessary to develop a new catalyst with good stability, high efficiency and easy recovery.

Poly(ionic liquid)s (PILs), also known as polymeric ionic liquids, are a subclass of polyelectrolytes characterized by the presence of ionic liquid (IL) monomers in each repeating unit, linked by polymer backbones to form a macromolecular structure. PILs combine the excellent property of ionic liquids and polymers, and the advantages of better mechanical properties, stability and processability, which are widely used in absorption, catalysis, electrochemistry and other fields [26–28]. Metal ion catalysts are known to be very effective in PET methanolysis, but have the drawback of being difficult to recovery, which can be well addressed by the low solubility of the polymer. Polymerization of one or more monomers can be attempted to achieve high activity and recyclability of the catalyst while retaining the active groups. Based on this idea, a variety of synthesized poly(ionic liquid)s were used as catalysts to achieve the recycling of PET. To obtain an efficient and recyclable non-heterogeneous catalyst for PET methanolysis, 1-vinyl-3-ethylimidazolium acetate ([VELm]Ac), whose branched chain contains unsaturated C=C bonds, was chosen to be copolymerized with metal salts containing unsaturated bonds, which are a key part of methanolysis.

Herein, poly(ionic liquid)s containing different metal ions were synthesized and characterized by FT-IR, XRD, TGA and ICP. The synthesized poly(ionic liquid)s were used to catalyze the methanolysis of PET. The effects of reaction temperature, reaction time, methanol amount, catalyst amount and sources of PET feedstock on the methanolysis reaction were investigated. Recovery experiments were conducted on poly(ionic liquid) catalysts, and the metal ion content of the initial catalyst and the catalyst after six repetitions was determined by ICP. The main objective of this work is to provide a recyclable, green and efficient catalyst and to facilitate the development of PET methanolysis.

2. Experimental

2.1. Materials

PET bottle chips (average size 5 mm × 5 mm) were purchased from Beijing Incom Resources Recovery Recycling Co., Ltd. PET colored fibers were obtained at Yiwu Najie ribbon Co., Ltd. Zinc acrylate ((CH₂=CHCOO)₂Zn, ZDAA) (98%), potassium acrylate (CH₂=CHCOOK, AAK) (95%), sodium acrylate (CH₂=CHCOONa, AANA) (95%), cobalt carbonate (CoCO₃, AR) and 2,2'-azobis(2-methylpropionitrile) (98%) were purchased from Shanghai Macklin Biochemical Co., Ltd. 1-Ethyl-3-vinylimidazole acetate ([VELm]Ac, 97%) was purchased from Shanghai Chengjie Chemical Co., Ltd. Methanol (AR), acetonitrile (AR) and acrylic acid (AAc, 99%) were obtained at Tianjin Damao Chemical Reagent Factory. Ammonium persulphate (ACS) and dimethyl terephthalate (Standard DMT, 99%) were obtained from Shanghai Aladdin Biochemical Technology Co., Ltd.

2.2. Synthesis of catalyst

2.2.1. Synthesis of Co(AAc)₂·H₂O precursor

Co(AAc)₂·H₂O precursor was synthesized according to the previous report [29], which was prepared by the reaction of CoCO₃ with AAc. CoCO₃ powder and AAc were mixed in water in a molar ratio of 2: 1. The mixture was stirred continuously at 50 °C for at least 48 h and then filtered while hot. The filtrate was vacuum distilled to remove some of the water and unreacted AAc, and then the residue was cooled to isolate the purple precipitate of Co(AAc)₂·H₂O. The resulting solid was then rinsed with a small amount of water and dried under vacuum at 60 °C for 24 h before use.

2.2.2. Synthesis of PIL-Co²⁺

In a flask, 1-ethyl-3-vinyl imidazole acetate (0.02 mol, 3.64 g), cobalt acrylate (0.01 mol, 2.01 g), 2,2'-azobis(2-methylpropionitrile) (5 wt%, 0.28 g) as initiator and acetonitrile as solvent were successively added, then heated and stirred at 70 °C for 12 h under the protection of N₂. After cooling, the sample was filtered and washed with distilled water and acetonitrile for three times, and then dried under vacuum, and finally a purple powder solid was obtained, which is PIL-Co²⁺.

2.2.3. Synthesis of PIL-Zn²⁺

PIL-Zn²⁺ was synthesized in a similar way to that of PIL-Co²⁺. Finally, light yellow powder solid was obtained as PIL-Zn²⁺.

2.2.4. Synthesis of PIL-Na⁺ and PIL-K⁺

The synthesis of PIL-Na⁺ and PIL-K⁺ is also similar to that of PIL-Zn²⁺. However, the solvent was water and the initiator was ammonium persulfate. After the reaction, a flocculent precipitate was obtained. Then the sample was filtered, and washed with distilled water and acetonitrile for three times, and then vacuum dried and ground to obtain solid powder, which are PIL-K⁺ and PIL-Na⁺.

2.3. Methanolysis of PET

An appropriate amount of PET, methanol and catalyst were loaded into a pressure bottle reactor with a magnetic stirrer and the reaction was stirred at the desired temperature (140–180 °C) for the desired reaction time (30–90 min). After the reaction, the reaction mixture was cooled to room temperature, and completely dissolved with a certain amount of acetonitrile, and then filtered and separated to obtain the catalyst, undegraded PET and oligomers. The residual PET was washed, dried at 70 °C for 12 h, then weighed to calculate the conversion of PET, which is defined by Eq. (2–1):

Conversion of PET

$$= \frac{\text{Initial weight of PET} - \text{Residual weight of PET}}{\text{Initial weight of PET}} \times 100\% \quad (2-1)$$

The yield of DMT was determined by quantitative analysis of HPLC (LC-20AT, Shimadzu, Japan) by external standard method. The analytical conditions were as follows: Shimadzu InertSustain AQ-C18 (5μm, Columns 4.6 × 250 mm) column, mobile phase of 70% methanol and 30% ultrapure water, flow rate of 0.5 ml·min⁻¹, and column temperature of 35 °C. The yield of DMT is calculated by Eq. (2–2):

DMT yield

$$= \frac{\text{Weight of DMT} / \text{Molecular weights of DMT}}{\text{Initial weight of PET} / \text{Molecular weights of PET repeat unit}} \times 100\% \quad (2-2)$$

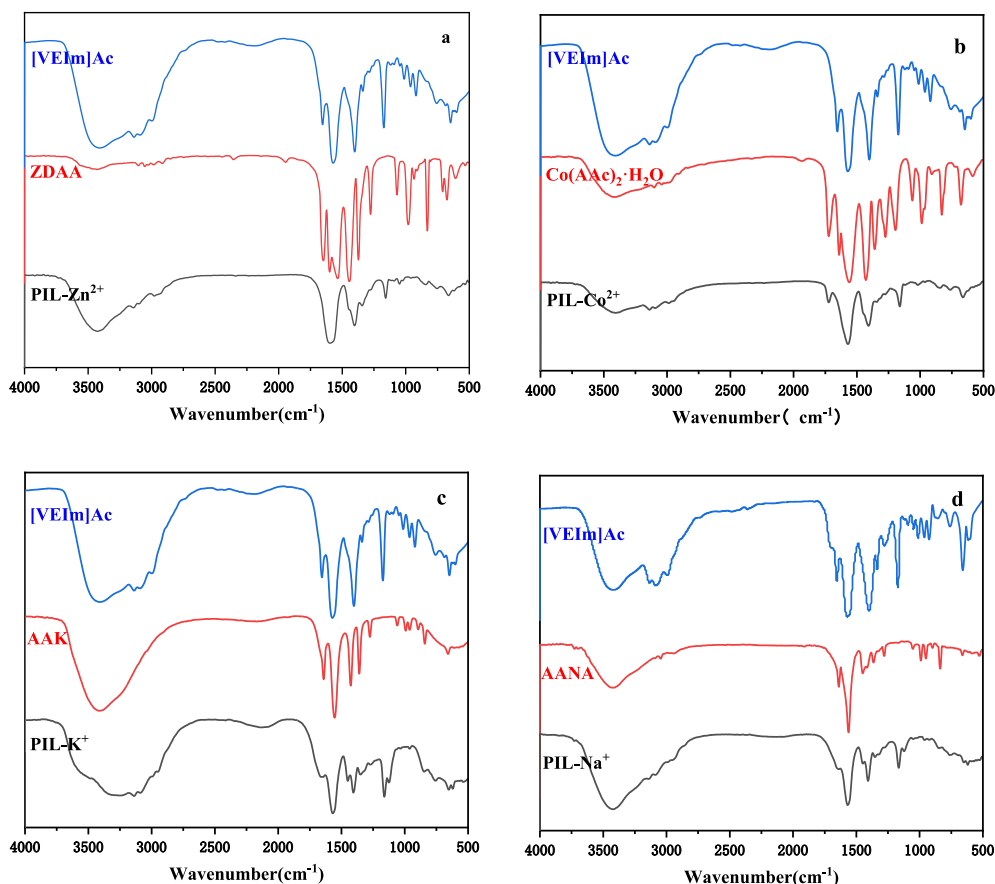


Fig. 1. (a) FT-IR spectra of [VEIm]Ac, ZDAA and PIL-Zn²⁺, (b) FT-IR spectra of [VEIm]Ac, Co(AAc)₂·H₂O and PIL-Co²⁺, (c) FT-IR spectra of [VEIm]Ac, AAK and PIL-K⁺, (d) FT-IR spectra of [VEIm]Ac, AANA and PIL-Na⁺.

2.4. Characterization and instruments

FT-IR spectra of the poly(ionic liquid)s and the main products were obtained with a Nicolet 380 (Thermo Fisher Scientific, USA) spectrometer. X-ray diffraction (XRD) patterns of the catalysts before and after recycling were recorded in the range $2\theta = 5\text{--}90^\circ$ using a Smartlab (9) (Rigaku, Japan) with CuK α radiation. The instrument was operated at 45 kV and 30 mA. The spectra were recorded at a scanning rate of $5^\circ/\text{min}$. Thermogravimetric analysis (TGA) curves and differential scanning calorimetric analysis (DSC) of raw materials, main product and poly(ionic liquid)s were obtained using TG-DTA6300 (Japan), heated from room temperature to 600°C at a rate of $10^\circ\text{C}/\text{min}$ under nitrogen atmosphere. Nuclear magnetic spectra (NMR) were measured by an ECA-600 NMR instrument (Bruker, Germany) with CDCl_3 as solvent. MS was obtained by electrospray ionization mass spectrometry (ESI-MS) on a UPLC-QTOF instrument (Bruker, Germany). The Zn^{2+} content of the catalyst was measured by ICPE-9000 (Shimadzu, Japan).

3. Result and discussion

3.1. Characterization of the synthesized poly(ionic liquid)s

PIL-Zn²⁺, PIL-Co²⁺, PIL-K⁺ and PIL-Na⁺ were analyzed using FT-IR, and the results are shown in Fig. 1(a-d). The bands at 1434 and 1600 cm^{-1} are separately assigned to the symmetric peak and asymmetric stretch peak of the carbonyl group in a acrylate molecule, which adequately indicate the coordination of acrylic and metal ions [29]. The $=\text{C-H}$ vibrational absorption peaks on the imidazole ring at 3141 and 2973 cm^{-1} , the C=N and aromatic back-

bone vibrational absorption peaks on the imidazole ring at 1598 and 1450 cm^{-1} , the stretching vibration of the acetate at 1598 and 1398 cm^{-1} , the C-N vibrational absorption peak on the imidazole ring at 1160 cm^{-1} , and the vibrational absorption peak of the imidazole ring at 1051 cm^{-1} . The absorption peak of monomer C=C at 1650 cm^{-1} , and the absorption peak of C=C disappears after polymerization.

3.2. Qualitative analysis of the main degradation product

To determine the structure of PET methanolysis product catalyzed by PILs, various characterization methods such as DSC, XRD, NMR, FT-IR and ESI-MS were performed. By analyzing the main product obtained from PET methanolysis and the purchased DMT standard (Fig. 2), it was found that the obtained product and the DMT standard had the same structure, thus the degradation product was identified as DMT. For example, the DSC analysis of the main product is shown in Fig. 2(a). It can be observed that a sharp absorption peak appears at 142°C , which is close to the known melting point of DMT standard (144°C) [30]. It is tentatively assumed that the main product is DMT. As shown in Fig. 2(b), the XRD patterns of the obtained products were compared with those of the DMT standard, and the structures were found to be consistent. In the ^1H NMR spectrum Fig. 2(c), the main peak at 3.87 ppm corresponds to the H on the methyl group ($-\text{CH}_3$) attached to the ester group ($\text{COO}-$). The single signal at 8.02 ppm indicates the presence of four aromatic protons of the benzene ring. The position around 7.33 ppm is the H of the solvent CDCl_3 Fig. 2(d) shows the FT-IR spectra of the main product, DMT spectra and PET raw material, and the spectrum for the obtained DMT was in good accor-

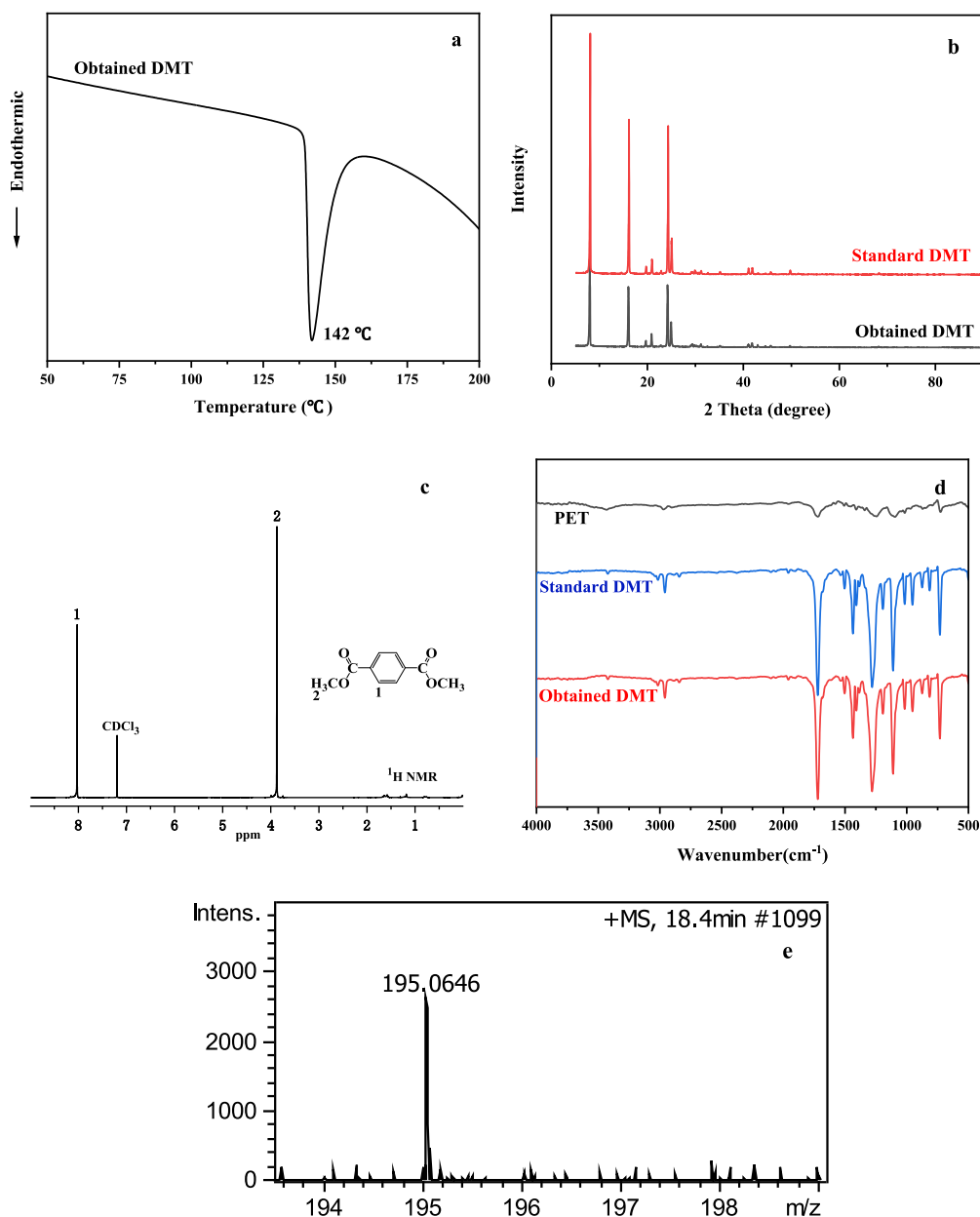


Fig. 2. (a) DSC curve of DMT, (b) XRD spectra of standard DMT and obtained DMT, (c) ^1H NMR spectrum of DMT, (d) FT-IR spectra of PET, standard DMT and obtained DMT, (e) ESI-MS spectrum of obtained DMT.

dance with that of the DMT standard. The FT-IR spectrum of main product is show that the typical bands at 3421 and 1135 cm^{-1} correspond to the hydroxyl fraction. Stretching band of $\text{C}=\text{O}$ appears at 1722 cm^{-1} , alkyl $\text{C}-\text{H}$ bands could be found at 2960 cm^{-1} , aromatic $\text{C}-\text{H}$ bands at 1434 and 1500 cm^{-1} , all of which are present in DMT. ESI-MS analysis was performed in Fig. 2(e), and the results showed that the peak with the highest intensity at $m/z = 195$, consistent with the molecular weight of the DMT ionized by H^+ . Therefore, the main product of PET methanolysis is DMT.

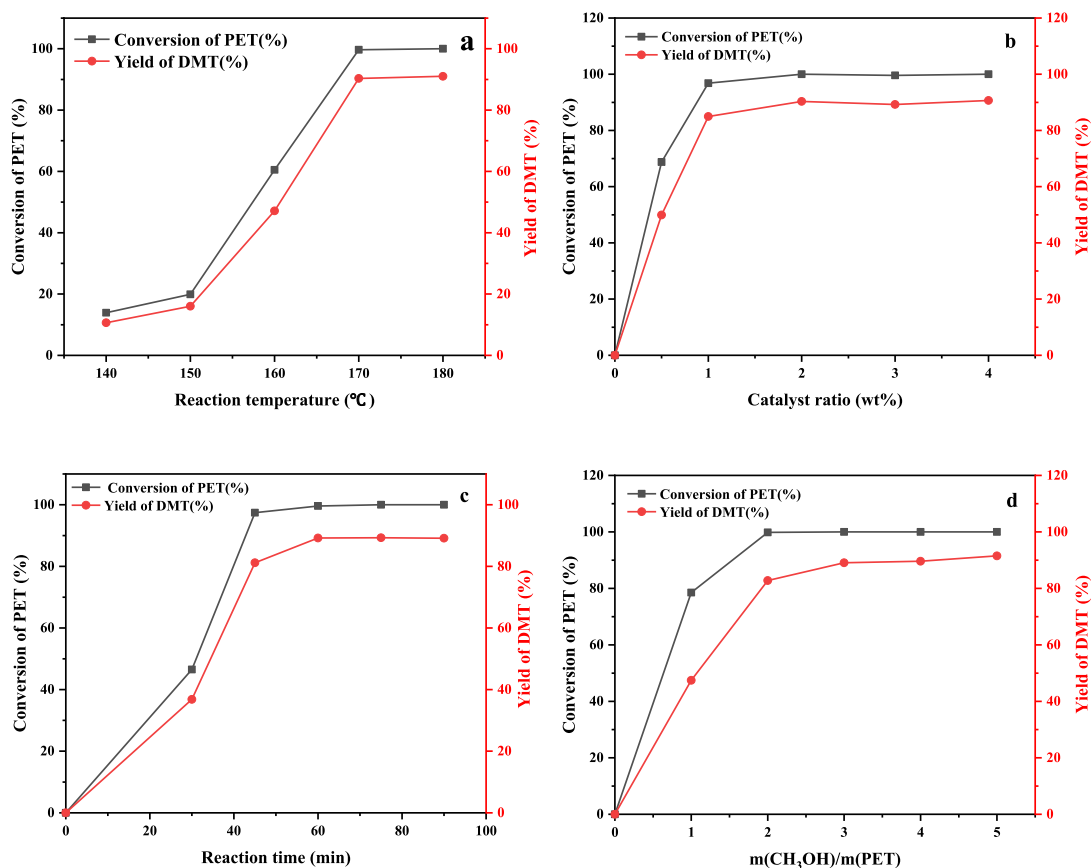
3.3. Selection of the catalysts

The catalytic performance of the prepared catalysts in PET methanol decomposition and the comparison with the catalyst performance in the literature were investigated. As shown in Table 1, no PET methanolysis occurred in the absence of catalyst or using $[\text{VEIm}]\text{Ac}$ as catalyst. However, when PIL-Zn^{2+} and PIL-Co^{2+} were

used, the PET conversion was the highest at 100% and the DMT yields were 90.3% and 69.2%, respectively. When PIL-Na^+ and PIL-K^+ were used, the conversion of PET were low with 4.9% and 5.9%, and the yields of the products DMT were 2.2% and 0.7%. This result indicates that PIL-Zn^{2+} has excellent catalytic activity for the methanolysis of PET compared to other metal-containing catalysts (Co, Na, K, Al). This finding coincides with the literature that zinc-based catalysts showed higher activity for PET depolymerization [25]. Kurokawa et al [31], used AIP as catalyst for the methanolysis of PET, but with significantly more methanol and catalyst dosage and higher reaction temperature ($200\text{ }^\circ\text{C}$). In addition, Pham and Cho [32] work although the reaction occurred at $25\text{ }^\circ\text{C}$, but also had the disadvantage of higher methanol and catalyst mass, longer reaction time (24 h) and the usage of hazardous solvent dichloromethane which was difficult to remove in the product. Therefore, PIL-Zn^{2+} , as the ideal catalyst, was chosen for further investigation.

Table 1
Methanolysis of PET using different catalysts.

Entry	MeOH: PET	Cat.	Mass ratio of cat./PET	Reaction Time (h)	Reaction temperature (°C)	Co-solvent	Conv. of PET (%)	Yield of DMT (%)	Ref.
1	4: 1	/	/	1	170	/	/	/	This work
2	4: 1	[VEIm]Ac	2%	1	170	/	/	/	This work
3	4: 1	PIL-Zn ²⁺	2%	1	170	/	100	90.3	This work
4	4: 1	PIL-Co ²⁺	2%	1	170	/	100	69.2	This work
5	4: 1	PIL-Na ⁺	2%	1	170	/	4.9	2.2	This work
6	4: 1	PIL-K ⁺	2%	1	170	/	5.9	0.7	This work
7	16: 1	AIP	10%	2	200	/	95.4	67.3	[31]
8	16: 1	AIP	10%	2	200	toluene	100	88.5	[31]
9	50: 1	K ₂ CO ₃	20%	24	25	CH ₂ Cl ₂	100	93.1	[32]

**Fig. 3.** Effects of different reaction parameters on PEF methanolysis: (a) reaction temperature (b) catalyst ratio (c) reaction time (d) m(CH₃OH)/m(PET). If not specified, the default reaction conditions were as follows: catalyst (2 wt% PET), reaction time (60 min), reaction temperature (170 °C), m (CH₃OH): m (PET) = 4: 1.

3.4. Effects of reaction conditions on the methanolysis of PET

3.4.1. Optimization of reaction conditions

The effect of temperature on the conversion of PET and the yield of DMT was investigated, the results are shown in Fig. 3(a). It can be concluded that the temperature plays a key role in the methanolysis of PET. When the reaction temperature was 140 and 150 °C, the conversions of PET were only 12.9% and 19.9%, and the yields of DMT were also very low. The conversion of PET was significantly increased when the reaction temperature was increased from 150 to 170 °C. At 170 °C, PET was completely degraded and the product DMT yield reached 90.3%. It has been suggested that higher reaction temperatures facilitate the contact between more feedstock molecules and the active sites of the catalyst. In addition, PET, as a highly crystalline polymer, has higher solubility in methanol at higher temperature, and the molecular chains in the crystalline region are easy to open and swell, thus fac-

ilitating the reaction [10]. When the temperature was finally increased to 180 °C, the yield of DMT remained almost constant at about 91.0%. This is because of the chemical reaction equilibrium between DMT monomer and 2-hydroxyethyl methyl terephthalate (MHET) by-products in the reaction system. However, the increase of temperature not only accelerates the rate of PET methanolysis, but also decreases the concentration of methanol in the reaction system. In addition, the concentration of the generated ethylene glycol (EG) in the reaction system becomes relatively larger, which causes the equilibrium to move toward MHET and BHET production [33]. From the perspective of practicality and energy saving, 170 °C is the best reaction temperature.

Fig. 3(b) represents the effect of catalyst amount on the methanolysis reaction. According to the blank experiment, PET was almost not converted (conversion of PET < 0.1%). As the amount of catalyst was increased from 0 to 1 wt%, PET was nearly completely converted and the yield of DMT rapidly increased to 85.0%. When

the amount of catalyst was further increased to 2 wt%, PET was completely converted and the yield of DMT reached the maximum equilibrium value of 90.3%. The increase in DMT yield with the increase in catalyst loading may be attributed to the increase in the number and availability of catalytically active sites [25]. However, the further increase in the amount of catalyst did not result in a significant increase in the yield of DMT. These results indicate that under these reaction conditions, 2 wt% of the active sites of the catalyst is sufficient for the PET degradation reaction. Therefore, the catalyst addition was chosen to be 2% of PET mass from the viewpoint of cost and post-treatment convenience.

The effect of reaction time on PET degradation and DMT yield was examined using PIL-Zn²⁺ catalyst (Fig. 3(c)). At the beginning of the methanolysis reaction, i.e., 30–60 min, both the conversion of PET and the yield of DMT increased rapidly with the increase of reaction time. In contrast, the conversion of PET and the yield of DMT remained almost constant in the range of 60–90 min. This is due to the low DMT concentration in the solution at the early stage of the reaction and the low mass transfer resistance of the products from PET surface, which facilitates the dispersion and dissolution of the products. In addition, PET methanolysis is an equilibrium reaction, thus, the lower concentration of DMT results the lower the rate of the reverse reaction. As the reaction proceeds, the concentration of DMT in the solution keeps increasing, the mass transfer resistance and the inverse reaction rate of DMT on the PET surface also keep increasing, therefore, the reaction rate will decrease until gradually reaching equilibrium [10].

Furthermore, as shown in Fig. 3(d), the effect of CH₃OH/PET mass ratio ($m(\text{CH}_3\text{OH})/m(\text{PET})$) on PET methanolysis reaction was also examined. When the amount of CH₃OH was increased from 1: 1 to 2: 1, the conversion of PET was increased from 78.5 to 100% and the yield of DMT was increased from 47.2 to 82.8%. This was due to the lower methanol concentration in the reaction system, which affected the equilibrium between DMT and MHET, with more by-products MHET produced. When the mass ratio of methanol to PET was further increased to 3: 1, 4: 1 and 5: 1, PET was completely converted and the yield of DMT increased to about 90%, which was due to the fact that the concentration of formed EG in the reaction system was relatively low after increasing the amount of methanol, and the equilibrium between DMT and MHET moved to the direction that benefit to the production of DMT. However, the further increase of methanol dosage resulted in no significant increase of the yield of DMT, mainly because the concentration of catalyst decreased with the further increase of methanol amount [34,35]. To obtain a high DMT yield, a CH₃OH/PET mass ratio of 3: 1 was optimal for this reaction.

3.4.2. Reusability of PIL-Zn²⁺

PIL-Zn²⁺ is insoluble in the reaction system, so the catalyst can be recovered by dissolving the degraded product in acetonitrile and filtering it. The experiments were repeated for the PIL-Zn²⁺ catalyst under optimal degradation conditions. As shown in Fig. 4(a), there was no significant activity loss after six catalyst repetitions, which indicates the persistent high activity of PIL-Zn²⁺ for the methanolysis of PET. In addition, the analysis of the PIL-Zn²⁺ catalysts before and after recycling was performed by TGA, XRD, FT-IR and ICP. The thermal stability of PIL-Zn²⁺ was analyzed by TGA, as shown in Fig. 4(b). The onset decomposition temperature of PIL-Zn²⁺ is 250 °C, which is higher than that of both monomers ZDAA and [VELm]Ac. These results indicate that the prepared PIL-Zn²⁺ catalyst is thermally stable in the methanolysis reaction range of 140–180 °C. Fig. 4(c) shows the FT-IR analyzes of the catalyst before and after cycling. The IR spectra of used catalyst was consistent with the fresh catalyst, indicating that there was no change of catalyst during the reaction. Among them, the absorption peaks at 1720 and 1280 cm⁻¹ are the vibrational absorption peaks of C=O

Table 2
ICP analysis of fresh and used catalyst.

Catalyst	Zn content (wt%)	
	First time	Sixth time
PIL-Zn ²⁺	12.5	10.1

Reaction conditions: $m(\text{CH}_3\text{OH}): m(\text{PET}) = 3: 1$, PIL-Zn²⁺ (2 wt% PET), 170 °C, 60 min.

and C–O–C on DMT, respectively, which are due to the incomplete cleaning of the catalyst during the catalyst recovery process, resulting in the small amount of DMT adsorbed on the catalyst surface. Since these two peaks of DMT are particularly sensitive in the IR spectrum, even small amounts of residue can lead to the appearance of significant absorption peaks. However, this result does not affect the spectrum of the catalyst. In addition, the XRD plots of the catalyst PIL-Zn²⁺ before and after recycling (Fig. 4(d)) also reflect the stability of the catalyst structure well. ICP analysis of the PIL-Zn²⁺ catalyst recovered after six cycles was performed as shown in Table 2. There was no significant change in the metal ion content after six repetitions, which indicates that the catalyst is stable and has potential applications. The above results indicate that the catalyst PIL-Zn²⁺ has good stability.

3.4.3. Effect of PET sources

The effect of PET source on its methanol degradation was further investigated under the optimized conditions screened above, i.e., $m(\text{CH}_3\text{OH}): m(\text{PET}) = 3: 1$, PIL-Zn²⁺ catalyst (2 wt% PET), reaction temperature of 170 °C and reaction time of 60 min. As shown in Table 3, the raw materials mainly included PET colored fibers, PET film containing organic silica, PET film containing acrylic acid, PET film containing polyurethane and the white bottle piece studied previously. The results indicate that PET from different sources and colors had no significant effect on the degradation reaction, the activity of catalyst did not change depending on the feedstock, and the catalyst had good homogeneity, which is consistent with the results observed by Yang et al [36].

3.5. Kinetics of the PET methanolysis

In kinetic studies of polymer depolymerization, the number of reaction levels is usually considered to be first-order [37]. Therefore, it was initially assumed that the methanolysis of PET in the presence of PIL-Zn²⁺ was controlled by the first order kinetic Eq. (3–1):

$$\frac{d(C_{\text{PET}})}{dt} = -kC_{\text{PET}} \quad (3-1)$$

where k is the rate constant of the reaction and C_{PET} is the concentration of PET at time t .

$$C_{\text{PET}} = C_{\text{PET}0}(1 - X) \quad (3-2)$$

where X is the conversion rate of PET. Then, Eq. (3–3) could be written as follows:

$$\frac{dX}{dt} = k(1 - X) \quad (3-3)$$

Eq. (3–3) was integrated over time to obtain Eq. (3–4):

$$\ln \frac{1}{1 - X} = kt \quad (3-4)$$

The effect of reaction temperature on the rate of PET methanolysis in the presence of PIL-Zn²⁺ is shown in Fig. 5(a), and the linear correlation coefficients (R^2) are all above 0.99, indicating that $\ln[1/(1-X)]$ was proportional to the reaction time at different temperatures and the process follows the first-order kinetic equation.

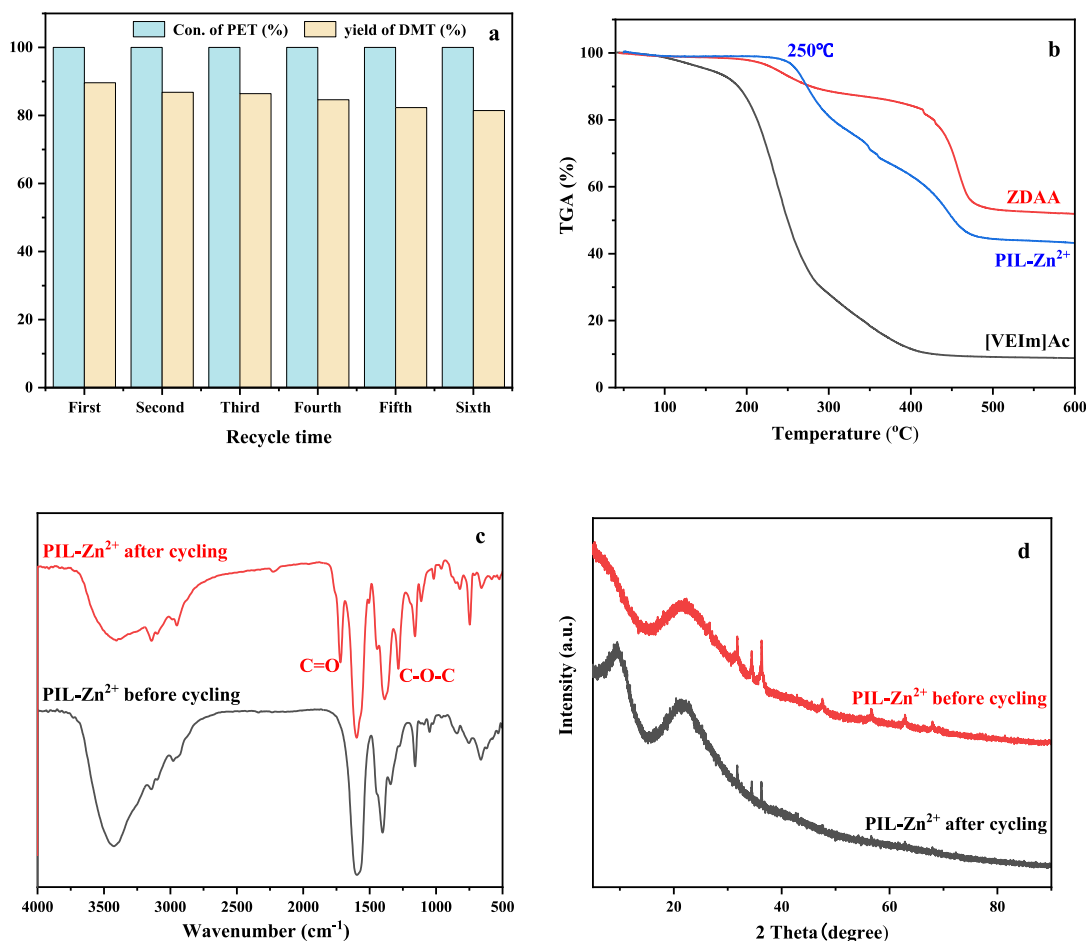


Fig. 4. (a) Catalytic recyclability, (b) TGA curves of ZDAA, [VEIm]Ac and PIL-Zn²⁺, (c) FT-IR spectra before and after PIL-Zn²⁺ cycling, (d) XRD spectra before and after PIL-Zn²⁺ cycling.

Table 3
Effect of PET source on the methanolysis of PET.

Entry	PET raw materials	Conv. of PET (%)	Yield of DMT (%)
1	White PET fiber	100	89.4
2	Black PET fiber	100	89.8
3	Blue PET fiber	100	89.1
4	Yellow PET fiber	100	90.1
5	Orange PET fiber	100	88.0
6	PET film sheets containing organic silica	100	85.6
7	PET film sheets containing polyurethane	100	78.9
8	PET film sheets containing acrylic acid	100	66.1
9	White bottle piece	100	89.1

Reaction conditions: $m(\text{CH}_3\text{OH}) : m(\text{PET}) = 3 : 1$, PIL-Zn²⁺ (2 wt% PET), 170 °C, 60 min.

The rate of PET methanolysis is positively correlated with the concentration of PET.

The slopes of the lines in Fig. 5(a) are 0.005, 0.012, 0.022 and 0.043 min⁻¹, respectively, which correspond to the rate constants of the methanolysis reactions at 140, 150, 160 and 170 °C, respectively.

Using these values of rate constants, the activation energy (E_a) is obtained by Eq. (3–5).

$$\ln k = \ln A - \frac{E_a}{RT} \quad (3-5)$$

where A is the pre-exponential factor, R is the gas constant (8.314 J·K⁻¹·mol⁻¹), and T is the temperature in Kelvin. Based on the slope of the Arrhenius plot shown in Fig. 5(b), the activation energy of the reaction was calculated to be 107.6 kJ/mol, which is significantly lower than previous value reported in the literature

for supercritical methanolysis of PET [30], which greatly reduces energy consumption.

3.6. Mechanism of the PET methanolysis

Based on the results of this study, a rational catalytic mechanism for PET methanolysis catalyzed by PIL-Zn²⁺ was proposed (Fig. 6). The positively charged Zn²⁺ in the catalyst attacks the C=O bond on the polyester, making the carbon cation more positively charged, while the negatively charged anion Ac⁻ interacts with the hydrogen in the hydroxyl group of methanol, making the oxygen in the methanol hydroxyl group more negatively charged and more likely to bind to the carbon cation next to the benzene ring [12,13]. The O on the methanol molecule joins with the C on the polyester C=O to form a new ester group and the

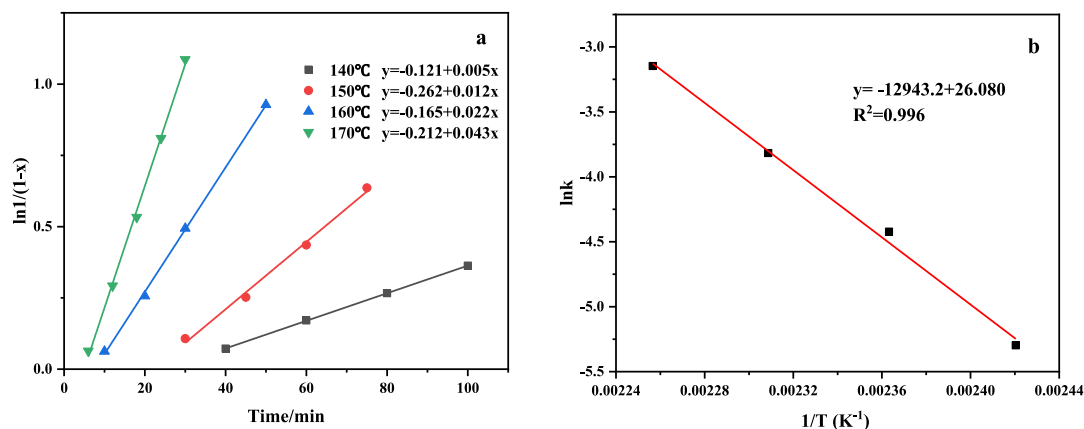


Fig. 5. (a) Effect of reaction temperature on the rate of methanolysis of PET in PIL-Zn²⁺, (b) Arrhenius plot of the rate constant of PET methanolysis in PIL-Zn²⁺.

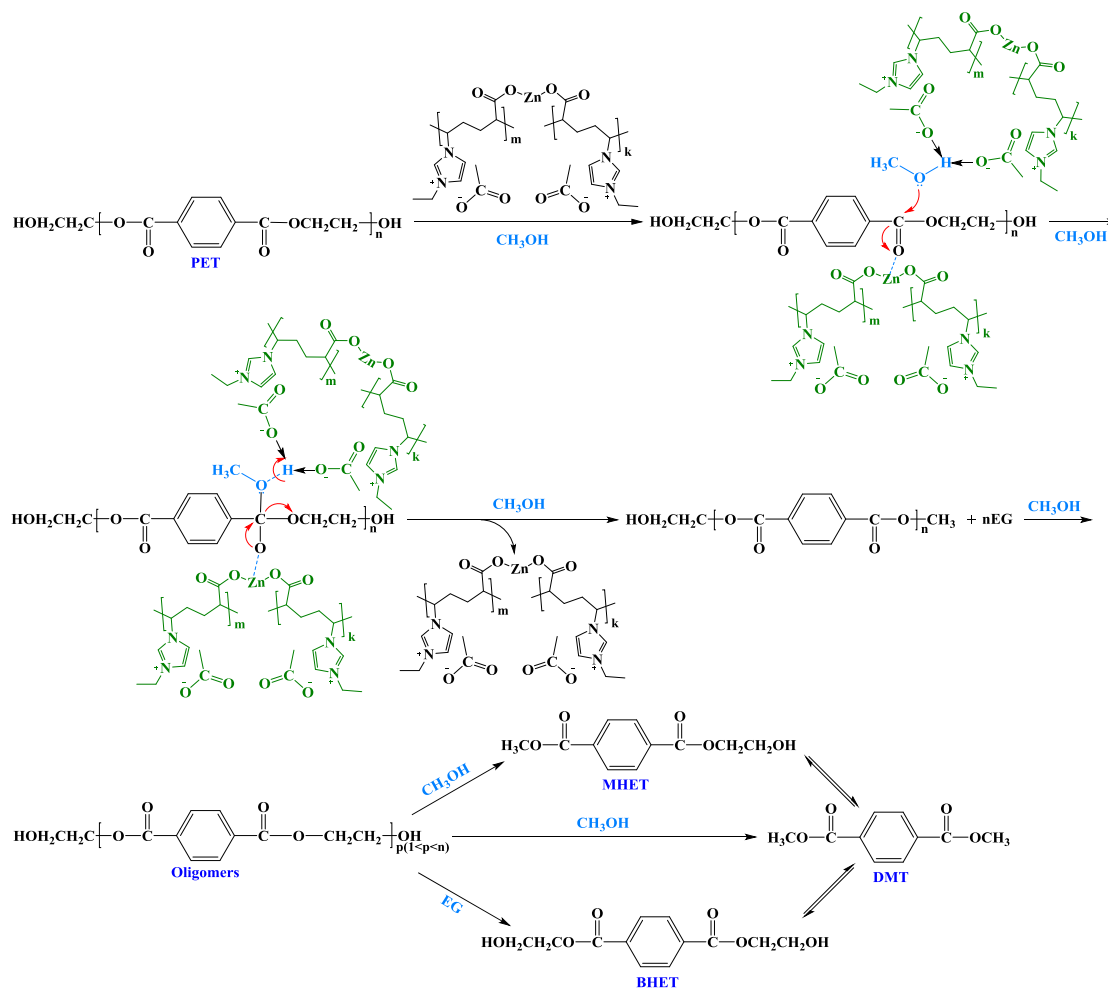


Fig. 6. Mechanism of the methanolysis of PET catalyzed by PIL-Zn²⁺.

polyester chain breaks. The polyester degraded into oligomers, dimers and monomers (BHET, MHET, DMT and EG). In the reaction system, BHET, MHET and DMT can transform into each other and eventually reach chemical equilibrium as the reaction proceeds [10].

4. Conclusion

In conclusion, PIL-Zn²⁺ catalyst can be used as an efficient and environmentally friendly catalyst for methanol degradation of PET.

The results showed that the reaction temperature is the key factor affecting the process. When the mass ratio of methanol to PET was 3: 1 and the amount of PIL-Zn²⁺ was 2% of PET mass, 100% PET conversion and 89.1% DMT yield could be achieved by methanolysis at 170 °C for 60 min. The PIL-Zn²⁺ catalyst was recycled six times without significant degradation in performance, and the poly(ionic liquid) has the potential to replace the conventional compounds used in industrial production for catalytic PET methanolysis with conventional compounds. In addition, the reaction kinetics were investigated and the results showed that the

methanolysis of PET is a first-order reaction with an activation energy of 107.6 kJ/mol.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Zhiqiang Jiang: Conceptualization, Methodology, Investigation, Writing – original draft, Visualization. **Dongxia Yan:** Supervision, Investigation, Writing – review & editing. **Jiayu Xin:** Methodology, Investigation, Writing – review & editing. **Fei Li:** Formal analysis. **Mengquan Guo:** Validation. **Qing Zhou:** Supervision, Conceptualization, Resources. **Junli Xu:** Supervision, Resources. **Yufeng Hu:** Writing – review & editing, Supervision. **Xingmei Lu:** Supervision, Conceptualization, Writing – review & editing.

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